



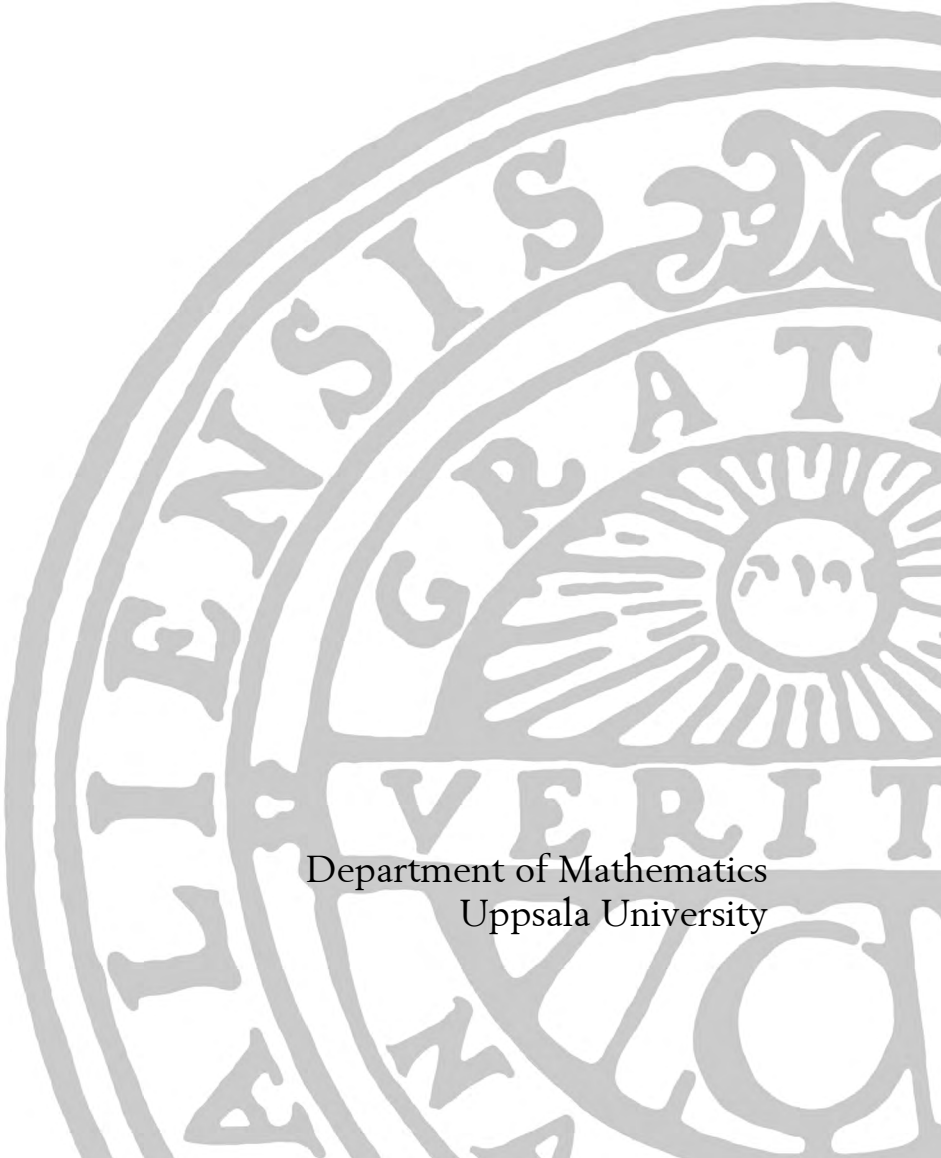
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Optimal Importance Sampling for Diffusion Processes

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Abstract

Variance reduction techniques are used to increase the precision of estimates in numerical calculations and more specifically in Monte Carlo simulations. This thesis focuses on a particular variance reduction technique, namely importance sampling, and applies it to diffusion processes with applications within the field of financial mathematics. Importance sampling attempts to reduce variance by changing the probability measure. The Girsanov theorem is used when changing measure for stochastic processes. However, a change of the probability measure gives a new drift coefficient for the underlying diffusion, which may lead to an increased computational cost. This issue is discussed and formulated as a stochastic optimal control problem and studied further by using the Hamilton-Jacobi-Bellman equation with a penalty term to account for computational costs. The objective of this thesis is to examine whether there is an optimal change of measure or an optimal new drift for a diffusion process. This thesis provides examples of optimal measure changes in cases where the set of possible measure changes is restricted but not penalized, as well as examples for unrestricted measure changes but with penalization.

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1 Introduction

Variance reduction techniques are used to increase the precision of estimates in numerical calculations and more specifically in Monte Carlo simulations. Some of the most commonly used techniques are antithetic variates, control variates and importance sampling. This thesis focuses on importance sampling and applies it to diffusion processes with applications primarily in the field of financial mathematics.

Importance sampling attempts to reduce variance by changing the probability measure. The Girsanov theorem is used when changing measure for stochastic processes. After applying importance sampling, the resulting diffusion process that will generate a smaller variance of estimation might have, for example, an exploding drift causing large fluctuations. To handle this numerically, smaller time steps in the Monte Carlo simulation might be needed to compensate the possible loss of precision. We arrive to a problem of finding a balance between the number of sample paths, the size of the time steps and the magnitude of the drift added to the diffusion process when using the Girsanov theorem to change measure in the importance sampling method. To generalize the situation, there are two types of estimation errors that occur in the method of Monte Carlo and importance sampling, namely non-negative variance inherent in the Monte Carlo approach and an error in connection with discretization of the stochastic differential equation.

To add a constant drift in the method of importance sampling is presumably not optimal. For the technique to be as efficient as possible, we need to allow for an exploding drift somewhere in time and space. However, non-constant (possibly exploding) coefficients lead to numerical difficulties when simulating trajectories. Indeed, such a situation calls for adaptive methods to distribute mesh points which are more complex than standard methods with time steps of a fixed size. We do not wish for the new and improved stochastic process with smaller variance to be difficult to simulate. The central dilemma of this thesis is consequently the trade-off between the improvements due to importance sampling and the numerical efficiency of the problem and its simulation.

Our next approach is to penalize these circumstances where additional computational costs arise after reducing the variance. This issue is discussed and formulated as a stochastic optimal control problem and studied further using the Hamilton-Jacobi-Bellman equation with a penalty term to account for computational cost. The objective of this thesis is thus to examine whether there is an optimal change of measure or an optimal new drift for a diffusion process. Further follows a discussion on the trade-off between the benefits of importance sampling and its computational costs.

In the following section, we introduce importance sampling and touch upon some of the numerical issues in Monte Carlo methods. In Section 3 and 4, we go through needed background material such as change of measure and the Girsanov theorem, but also stochastic optimal control problem and the Hamilton-Jacobi-Bellman equation. In Section 5 and 6, we apply these theorems to carefully selected examples.

2 Monte Carlo Methods

Monte Carlo methods are a class of computational algorithms that can be applied to vast ranges of problems. They provide approximate solutions and are used in cases where analytical or numerical solutions do not exist or are too difficult to implement. It is a computational algorithm that makes numerical estimations by taking the empirical mean of repeated random sampling. It is an easy way of modeling complex situations which allows for applications in a wide range of fields such as finance and engineering.

When simulating Monte Carlo methods there are two main factors that affect the cost-effectiveness: the number of sample paths and the size of the time steps. Let N be the number of sample paths and $h = \Delta t$ be the size of the time steps in the general Monte Carlo integration method. Then, according to Seydel (2009) and Hirs (2012), the numerical errors or rates of convergence depending on h and N are

$$\epsilon_h = \mathcal{O}(\sqrt{h}),$$

respectively,

$$\epsilon_N = \mathcal{O}(1/\sqrt{N}).$$

Further explanations of these rates of convergence are found in Section 2.1 and 2.3.

2.1 Monte Carlo Integration and Convergence of Error

Assume a probability distribution with density f , then the expectation of a function h is

$$\mathbb{E}[h(X)] = \int_{\mathbb{R}} h(x)f(x)dx.$$

In the one-dimensional case, for a definite integral on some interval $\mathcal{I} = [a, b]$, we use the uniform distribution with density

$$f = \frac{1}{b-a} \mathbb{1}_{\mathcal{I}} = \frac{1}{d(\mathcal{I})} \mathbb{1}_{\mathcal{I}},$$

where $d(\mathcal{I})$ denotes the length of the interval $\mathcal{I}$. Let

$$\alpha := d(\mathcal{I})\mathbb{E}[h(X)] = \int_a^b h(x)dx.$$

For independent samples $X_i \sim \mathcal{U}[a, b]$, the law of large numbers implies that the approximation

$$\hat{\alpha}_N := d(\mathcal{I})\frac{1}{N} \sum_{i=1}^N h(X_i) \rightarrow \alpha \text{ a.s. as } N \rightarrow \infty.$$

To generalize for the higher dimensional case, let $\mathcal{I} \subset \mathbb{R}^m$. We want to calculate the integral

$$\alpha^m := \int_{\mathcal{I}} h(x)dx.$$

Again, we draw independent and uniformly distributed samples $X_1, \dots, X_N \in \mathcal{I}$, then we get the approximation

$$\hat{\alpha}_N^m := d_m(\mathcal{I}) \frac{1}{N} \sum_{i=1}^N h(X_i),$$

where $d_m(\mathcal{I}) < \infty$ now is the volume, or the m -dimensional Lebesgue measure, of $\mathcal{I}$. Following the law of large numbers, $\hat{\alpha}_N^m$ converges almost surely to $\alpha^m = d_m(\mathcal{I})\mathbb{E}[h(X)] = \int_{\mathcal{I}} h(x)dx$ as $N \rightarrow \infty$. Let

$$\delta_N := \int_{\mathcal{I}} h(x)dx - \hat{\alpha}_N^m$$

be the error. Before deriving the variance of the error, let us examine the zero mean and correlation properties. We have

$$\begin{aligned} \delta_N &= \int_{\mathcal{I}} h(x)dx - d_m(\mathcal{I}) \frac{1}{N} \sum_{i=1}^N h(X_i) \\ &= \frac{1}{N} \sum_{i=1}^N \left(\int_{\mathcal{I}} h(x)dx - d_m(\mathcal{I})h(X_i) \right) \\ &= \frac{1}{N} \sum_{i=1}^N \left(\int_{\mathcal{I}} h(x) \frac{1}{d_m(\mathcal{I})} dx - h(X_i) \right) d_m(\mathcal{I}) \\ &= \frac{d_m(\mathcal{I})}{N} \sum_{i=1}^N \left(\int_{\mathcal{I}} h(x) \frac{1}{d_m(\mathcal{I})} dx - h(X_i) \right). \end{aligned}$$

It is easy to show that $\int_{\mathcal{I}} h(x) \frac{1}{d_m(\mathcal{I})} dx - h(X_i)$ has zero mean and considering X_i and X_j are independent for $i \neq j$, then $\int_{\mathcal{I}} h(x) \frac{1}{d_m(\mathcal{I})} dx - h(X_i)$ and $\int_{\mathcal{I}} h(x) \frac{1}{d_m(\mathcal{I})} dx - h(X_j)$ are uncorrelated. We have

$$\mathbb{E} \left[\int_{\mathcal{I}} h(x) \frac{1}{d_m(\mathcal{I})} dx - h(X_i) \right] = 0$$

and

$$\begin{aligned} &\mathbb{E} \left[\left(\int_{\mathcal{I}} h(x) \frac{1}{d_m(\mathcal{I})} dx - h(X_i) \right) \left(\int_{\mathcal{I}} h(x) \frac{1}{d_m(\mathcal{I})} dx - h(X_j) \right) \right] = \\ &\mathbb{E} \left[\int_{\mathcal{I}} h(x) \frac{1}{d_m(\mathcal{I})} dx - h(X_i) \right] \mathbb{E} \left[\int_{\mathcal{I}} h(x) \frac{1}{d_m(\mathcal{I})} dx - h(X_j) \right] = 0. \end{aligned}$$

Further, we can have a look at the variance of the error

$$\begin{aligned} \text{Var}(\delta_N) &= \mathbb{E}[\delta_N^2] - (\mathbb{E}[\delta_N])^2 \\ &= \mathbb{E}[\delta_N^2] \\ &= \frac{(d_m(\mathcal{I}))^2}{N^2} \sum_{i=1}^N \mathbb{E} \left[\left(\int_{\mathcal{I}} h(x) \frac{1}{d_m(\mathcal{I})} dx - h(X_i) \right)^2 \right] \\ &= \frac{(d_m(\mathcal{I}))^2}{N} \text{Var}(h), \end{aligned}$$

where the variance of h is

$$\text{Var}(h) := \int_{\mathcal{I}} h^2(x) \frac{1}{d_m(\mathcal{I})} dx - \left(\int_{\mathcal{I}} h(x) \frac{1}{d_m(\mathcal{I})} dx \right)^2.$$

Thus, the standard deviation of the error δ_N tends to zero with the order

$$\epsilon_N := \sqrt{\text{Var}(\delta_N)} = \mathcal{O}(1/\sqrt{N}).$$

Square integrability of h suffices ($h \in \mathcal{L}^2$), the integrands h need not be smooth (Seydel, 2009). Note that the error only depends on N . This implies that Monte Carlo resolves the curse of dimensionality, since the order $\epsilon_N = \mathcal{O}(1/\sqrt{N})$ of the error does not depend on the number of dimensions m (Hirsa, 2012).

The curse of dimensionality refers to the phenomena that as the number of dimensions or other features grows, the amount of data that needs to be analyzed grows exponentially. The conventional method for solving partial- or stochastic differential equations numerically is to discretize the continuous variables in space and time and solve the equation in discrete form. An example on how to do this can be found in Section 2.3. When using for example the method of finite differences, every dimension must be discretized and the number of discrete points where a solution has to be calculated increases exponentially with the number of dimensions. When instead using the Monte Carlo method, the amount of computational work grows only linearly with the number of dimensions. A disadvantage of Monte Carlo methods is instead that they generally offer slow convergence, requiring a very large number of simulations to yield a sufficiently accurate result.

2.2 Importance Sampling

Importance sampling is a method used to increase the efficiency of Monte Carlo simulations by reducing the variance of estimates. It does so by changing the probability measure from which paths are generated to increase sampling efficiency by giving more weight to "important" outcomes.

Say we want to Monte Carlo estimate an integral

$$\alpha := \mathbb{E}[h(X)] = \int_{\mathbb{R}^d} h(x) f(x) dx$$

where X is an $\mathbb{R}^d$ -valued random variable, h is a Borel function from $\mathbb{R}^d$ to $\mathbb{R}$, and $f(x)$ is the probability density function of X . The usual Monte Carlo estimator is

$$\hat{\alpha} := \hat{\alpha}(N) = \frac{1}{N} \sum_{i=1}^N h(X_i)$$

where X_i are i.i.d $\sim f$. Let g be another probability density function on $\mathbb{R}$ such that for all $x \in \mathbb{R}$, $f(x) > 0$ implies $g(x) > 0$. Now, if $\tilde{X}_i$ are independent draws from

the importance sampling distribution g , we can represent α as an expectation with respect to the density g

$$\alpha = \tilde{\mathbb{E}} \left[h(\tilde{X}) \frac{f(\tilde{X})}{g(\tilde{X})} \right] = \int_{\mathbb{R}^d} h(x) \frac{f(x)}{g(x)} g(x) dx. \quad (1)$$

Define the importance sampling estimator as

$$\hat{\alpha}_g := \hat{\alpha}_g(N) = \frac{1}{N} \sum_{i=1}^N h(\tilde{X}_i) \frac{f(\tilde{X}_i)}{g(\tilde{X}_i)}$$

where $\tilde{X}_i$ are independent draws from the importance sampling distribution g . It follows from equation (1) that $\hat{\alpha}_g$ is an unbiased estimator of α . The weight $\frac{f(\tilde{X}_i)}{g(\tilde{X}_i)}$ is called the *likelihood ratio* or *Radon-Nikodym derivative* evaluated at $\tilde{X}_i$.

Looking at the second moment

$$\begin{aligned} \tilde{\mathbb{E}} \left[\left(h(\tilde{X}) \frac{f(\tilde{X})}{g(\tilde{X})} \right)^2 \right] &= \int_{\mathbb{R}^d} h^2(x) \frac{f^2(x)}{g^2(x)} g(x) dx \\ &= \int_{\mathbb{R}^d} h^2(x) \frac{f(x)}{g(x)} f(x) dx \\ &= \mathbb{E} \left[h^2(X) \frac{f(X)}{g(X)} \right] \end{aligned}$$

with $\tilde{X} \sim g$ and $X \sim f$, we get the following variances

$$\text{Var}(\hat{\alpha}) = \frac{1}{N} (\mathbb{E} [h^2(X)] - \alpha^2)$$

and

$$\text{Var}(\hat{\alpha}_g) = \frac{1}{N} \left(\mathbb{E} \left[h^2(X) \frac{f(X)}{g(X)} \right] - \alpha^2 \right).$$

For importance sampling to be successful, it is crucial to find an effective importance sampling density g to reduce the variance (Glasserman, 2003).

The convergence rate depending on the number of Monte Carlo simulations N is the same after applying importance sampling as for the basic Monte Carlo method, i.e., $\mathcal{O}(1/\sqrt{N})$. It is possible to improve the speed of convergence by choosing a suitable importance sampling distribution g , but the overall convergence rate will behave the same disregarding a constant. The law of large numbers guarantees the convergence in both the general Monte Carlo case and when using importance sampling. If also the second moment

$$\int_{\mathbb{R}^d} h^2(x) \frac{f^2(x)}{g(x)} dx < \infty,$$

then the central limit theorem applies in the same way as before, and once again we get $\mathcal{O}(1/\sqrt{N})$. For more about importance sampling and convergence, see (Newton, 1997).

Example 2.1. Let

$$h(x) = \begin{cases} 1 & \text{if } x \geq A, \\ 0 & \text{if } x < A \end{cases}$$

for some $A \geq 0$. Further, let $X \sim N(0, 1)$ under P and $X \sim N(A, 1)$ under $\tilde{P}$. Then

$$\alpha = \mathbb{E}[h(X)] = \int_{\mathbb{R}} h(x)f(x)dx = \int_A^{\infty} f(x)dx,$$

where $f(x)$ is the $N(0, 1)$ -probability density function. Now,

$$\text{Var}(\hat{\alpha}) = \frac{1}{N} (\mathbb{E}[h^2(X)] - \alpha^2) = \frac{1}{N} ((1 - \Phi(A)) - (1 - \Phi(A))^2)$$

and

$$\text{Var}(\hat{\alpha}_g) = \frac{1}{N} \left(\mathbb{E} \left[h^2(X) \frac{f(X)}{g(X)} \right] - \alpha^2 \right)$$

where $\hat{\alpha}_g$ is the estimated value of α under $\tilde{P}$, with $g(x)$ denoting the $N(A, 1)$ -probability density function. With

$$\frac{f(x)}{g(x)} = \frac{\frac{1}{\sqrt{2\pi}}e^{-x^2/2}}{\frac{1}{\sqrt{2\pi}}e^{-(x-A)^2/2}} = \frac{e^{-x^2/2}}{e^{-(x^2+A^2-2Ax)/2}} = e^{(A^2-2Ax)/2},$$

we have

$$\begin{aligned} \text{Var}(\hat{\alpha}_g) &= \frac{1}{N} \left(\mathbb{E} \left[\mathbb{1}_{\{X \geq A\}} e^{(A^2-2Ax)/2} \right] - \alpha^2 \right) \\ &= \frac{1}{N} \left(\int_A^{\infty} \frac{e^{-x^2/2}}{\sqrt{2\pi}} e^{(A^2-2Ax)/2} dx - \alpha^2 \right) \\ &= \frac{1}{N} \left(\int_A^{\infty} \frac{e^{A^2}}{\sqrt{2\pi}} e^{-(x+A)^2/2} dx - \alpha^2 \right) \\ &= \frac{1}{N} \left(e^{A^2} \int_{2A}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy - \alpha^2 \right) \\ &= \frac{1}{N} \left(e^{A^2} (1 - \Phi(2A)) - (1 - \Phi(A))^2 \right). \end{aligned} \tag{2}$$

For this to be a successful change of measure in the method of importance sampling, we need to show that

$$e^{A^2} (1 - \Phi(2A)) \leq (1 - \Phi(A))$$

which would imply

$$\text{Var}(\hat{\alpha}_g) \leq \text{Var}(\hat{\alpha}).$$

From equation (2), we have

$$e^{A^2} (1 - \Phi(2A)) = \int_A^{\infty} \frac{e^{-x^2/2}}{\sqrt{2\pi}} e^{\frac{A^2}{2}-Ax} dx \leq \int_A^{\infty} \frac{e^{-x^2/2}}{\sqrt{2\pi}} dx = (1 - \Phi(A)).$$

The inequality can be explained by looking at the factor $e^{\frac{A^2}{2}-Ax}$. For $x \in [A, \infty)$,

$$\frac{A^2}{2} - Ax \leq 0$$

which implies that

$$0 \leq e^{\frac{A^2}{2}-Ax} \leq 1.$$

Since the probability density function is non-negative for every x , the $e^{\frac{A^2}{2}-Ax}$ -factor entails

$$e^{A^2}(1 - \Phi(2A)) \leq (1 - \Phi(A)).$$

Thus, the estimated value of α will have a smaller or equal variance under the new probability measure $\tilde{P}$.

2.3 Time Discretization Error

In Section 2.1, we found that the error depending on the number of Monte Carlo simulations converges according to $\mathcal{O}(1/\sqrt{N})$. In this section we will show that the error depending on the size of the time steps $h = \Delta t$ follows $\mathcal{O}(\sqrt{h})$.

To study the accuracy of numerical approximations depending on the size of the time steps in the general Monte Carlo integration method, we let X_t be a stochastic process and solution of the stochastic differential equation (SDE)

$$dX_t = \mu(X_t)dt + \sigma(X_t)dW_t, \text{ for } 0 \leq t \leq T,$$

with initial value X_0 for $t = 0$, where W is a Brownian motion.

When Monte Carlo simulating this stochastic process, we get a sample path X_t for each realization of the Brownian motion W_t . At each time step, we update the numerical approximation of the SDE and at the final step time we take the average of these sample paths to get an estimation of the value. Let

$$\epsilon_h := \mathbb{E} [X_T - Y_T^h] \tag{3}$$

be the error at time T , where Y_T^h is the approximation at T depending on the chosen step length h . One may perform numerical tests to see that the discretization error of a SDE when using Eulers method converges according to

$$\epsilon_h = \mathcal{O}(\sqrt{h}).$$

For the sake of simplicity, let the stochastic process X_t follow a geometric Brownian motion and thus satisfy the following stochastic differential equation

$$dX_t = \alpha X_t dt + \beta X_t dW_t, \text{ for } 0 \leq t \leq T. \tag{4}$$

We choose a geometric Brownian motion since it has an exact solution of which we can compare the approximations to. Using Euler discretization, the solution of a discrete version of the SDE in equation (4) denoted Y_{t_j} is

$$\begin{cases} Y_{t_{j+1}} = Y_{t_j} + \alpha Y_{t_j} \Delta t + \beta Y_{t_j} \Delta W_j, & t_j = j \Delta t, \\ \Delta W_j = W_{t_{j+1}} - W_{t_j} = Z \sqrt{\Delta t} \text{ with } Z \sim N(0, 1), \end{cases}$$

with initial value $Y_0 = X_0$. This discretization scheme can be used for more general SDEs as well. The step length $h = \Delta t$ is assumed equidistant. For $\Delta t = T/m$ the index j runs from 0 to $m - 1$. When $\beta \equiv 0$ we have a deterministic case and the discretization error of Euler's method of the ordinary differential equation (ODE) is $\mathcal{O}(h)$. With the error at time T defined in equation (3), one may perform numerical tests comparing the results from the Euler scheme to the exact value to see that the discretization error of a SDE when using Euler's method decreases slower than in the deterministic case. In fact, $\epsilon_h = \mathcal{O}(\sqrt{h})$.

Remark. With constant coefficients α and β , the analytical solution to the SDE in equation (4) is known, namely

$$X_t = X_0 e^{(\alpha - \frac{1}{2}\beta^2)t + \beta W_t}. \quad (5)$$

When there is an analytical solution to a SDE and if one is interested in the distribution of the process at a given fixed time (as opposed to a path-dependent quantity), there is no need to simulate the process for each time step. In this case, it is more effective to only take one single time step, since we know the analytical solution and the distribution of a Brownian motion. Although, it is not always the case that a SDE has a known solution, then you will need to discretize according to the Euler scheme to get a numerical solution. The Euler scheme can be generalized to other SDEs as well. There are also other discretization schemes that can be used, for example the Milstein method and the Runge-Kutta method. You may read more about these in Hirsä (2012).

However, since our objective is to derive methods that hold for large classes of stochastic differential equations (including SDEs with no explicit solution), we will not use the explicit form in equation (5), but instead use the Euler scheme. Furthermore, in some problems studied in later sections, the random variables depend on the whole path and not only on the value at one deterministic time. The reason for us to specify a model with explicit solution is that we then can perform exact calculations for comparison.

Below, a geometric Brownian motion has been simulated. As seen in Figure 1, the Euler approximation with smaller step size (b) is closer to the exact solution than the approximation with larger steps (a). Here we have simulated the SDE with $\alpha = 2$, $\beta = 1$ and $X_0 = 1$. What is referred to as the exact solution is a discretized Brownian path over the interval $[0, 1]$ with $\Delta t = 2^{-9}$ using the analytical solution from equation (5) as recommended in PC-Exercise 4.4.1 (Kloeden & Platen, 1992).

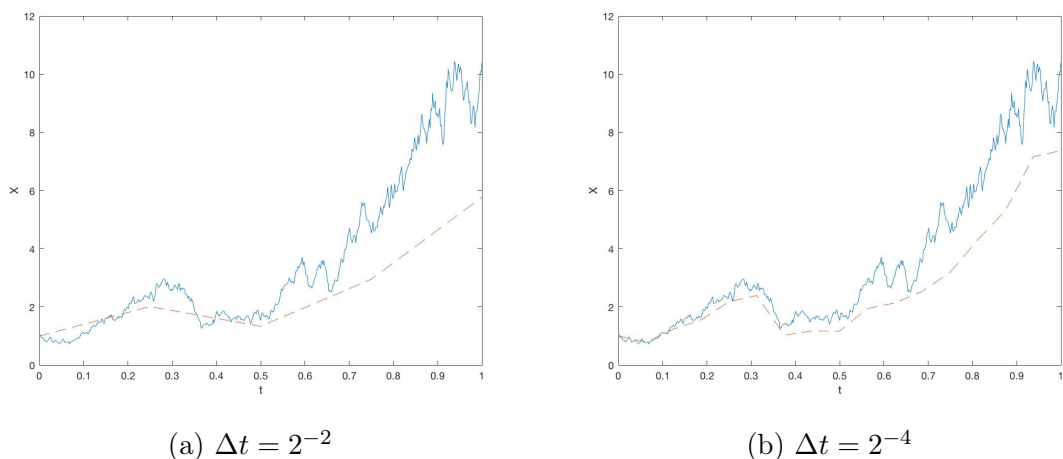


Figure 1: Euler approximation (dashed line) and exact solution

When simulating the Euler approximation for different time step sizes, one may expect a closer resemblance to the exact solution when using a smaller step size. To see that the error behaves according to $\epsilon_h = \mathcal{O}(\sqrt{h})$, one may compare the error as defined in equation (3) of different approximations Y_T^h for varying step lengths h . These simulations will show that the step length has a definite effect on the magnitude of the error which is proportional to the square root of the step size. To see it even more clearly, one may plot the result in log-log scale where the error will be a straight line with slope $\frac{1}{2}$, see Chapter 9, Section 3 in (Kloeden & Platen, 1992).

Below follows two examples on European call options with the underlying stock following a geometric Brownian motion. This is to illustrate the convergence of the error in Monte Carlo simulations. The definition of European call options is standard, see for example Björk (2009) where you may read further about this.

Definition 2.1 (European call option). *A European call option on the amount X with strike price K and exercise date T is a contract written at time $t = 0$ where the holder of the contract has the right, but not the obligation, to buy the amount X at the price K at time $t = T$.*

The contract function for a European call option is defined by

$$\Phi(x) = \max[x - K, 0],$$

and the price at time T is

$$\Pi(T) = \max[S(T) - K, 0],$$

where $S(t)$ denotes the underlying stock.

Proposition 2.1. *The price of a European call option with strike price K and time of maturity T is given by the Black-Scholes formula $\Pi(t) = F(t, S(t))$, where*

$$F(t, s) = sN(d_1(t, s)) - e^{-r(T-t)}KN(d_2(t, s)).$$

Here N denotes the cumulative distribution function for the $N(0, 1)$ distribution and is given by $N(x) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^x e^{-z^2/2} dz$, and

$$d_1(t, s) = \frac{1}{\sigma\sqrt{T-t}} \left[\ln \frac{s}{K} + \left(r + \frac{\sigma^2}{2} \right) (T-t) \right],$$

$$d_2(t, s) = d_1(t, s) - \sigma\sqrt{T-t}.$$

Example 2.2. A European call option where the underlying stock follows the dynamics

$$dS_t = rS_t dt + \sigma S_t dW_t$$

has been implemented using Euler's method. After Monte Carlo simulations with $r = 0.1$, $\sigma = 0.25$, the strike price $K = 15$, the initial point $S_0 = 14$ and the time of maturity $T = 0.5$, we can see in the following plots that the absolute value of the difference between the approximated value and the exact value of the option at time of maturity behaves as expected. The exact value of a European call option at time t is given by the Black-Scholes formula $\Pi(t) = F(t, S(t))$ from Proposition 2.1. Here, N and Δt denotes the number of Monte Carlo simulations respectively the size of the time steps.

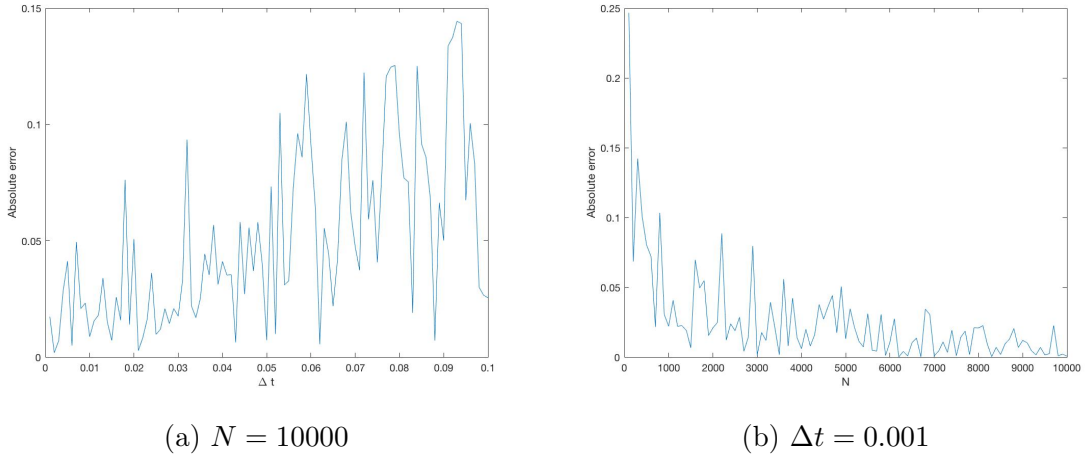


Figure 2: Error of European Call

Example 2.3. Given the same set up as for the European option in Example 2.2, say we have a given number of random numbers. These random numbers need to be drawn at each time step for each Monte Carlo simulation. How do we most wisely spend these random numbers?

Let the time of maturity be $T = 1$. In Figure 3a, the x-axis denotes the number of simulations N and in reversed order the number of sample paths, which is not shown on the axis. We are given 10000 random numbers. For N simulations, the number of sample paths is $\frac{10000}{N}$, and the size of the time step is $\Delta t = \frac{T}{10000/N}$. The figure shows that there are some ways of spending your random numbers that are more effective than others. In this example it seems like the most effective way to spend

your random numbers is somewhere around the usage of 400 time steps and 25 Monte Carlo simulations.

Example 2.4. To extend the previous problem in Example 2.3 to a more visually appealing graph that does not have to consider the divisibility of the number of random numbers, we extend the limit of random draws to the interval $[8000, 12000]$ instead of only allowing exactly 10000 random numbers. Now, let us iterate through an array of $N = 10, 20, 30, \dots, 3000$ simulations and divide T into $\frac{10000}{N}$ (rounded to the closest integer) time steps. The result is shown in Figure 3b. You can see that even though you take very small time steps, you will not get a good estimation if you have too few Monte Carlo simulations. The error seems to be even more drastic for a large number of Monte Carlo simulations if you take too few time steps. The magnitude of the error is large for N close to 3000 and also for small N , but in a more fluctuating manner. The relationship to the number of time steps is reversed. Again, this graph depicts the trade-off between the approximation error and the discretization error.

Remark. The Matlab function $Y = \text{round}(X)$ rounds each element of X to the nearest integer. In the case where an element has a fractional part of exactly 0.5, the round function rounds to the integer with larger absolute value.

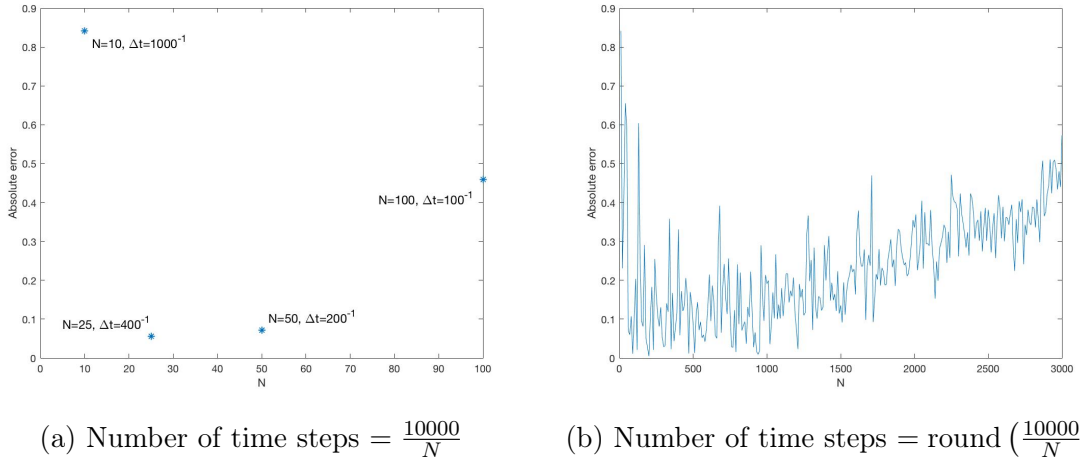


Figure 3: Error of European Call depending on both Δt and N

Remark. Note that this example shows large absolute errors due to the maximum number of simulations and the maximum number of time steps being rather small. This is of course for short simulation time purposes. In reality, one would devote more computational work for a more accurate result.

2.3.1 Smooth Coefficients

The following argument in the case of smooth coefficients is given by Carlsson, Moon, Szepessy, Tempone and Zouraris (2019). If we assume α, β and g are differentiable

to any order and these derivatives are bounded, then

$$\mathbb{E} [g(X_T) - g(Y_T^h)] = \mathcal{O}(h). \quad (6)$$

The Euler discretization Y of X can be extended for theoretical use to all t by

$$Y_t - Y_{t_j} = \int_{t_j}^t \bar{\alpha}(s, Y) ds + \int_{t_j}^t \bar{\beta}(s, Y) dW(s), \quad t_j \leq t \leq t_{j+1}$$

where, for $t_j \leq s \leq t_{j+1}$,

$$\begin{aligned} \bar{\alpha}(s, Y) &= \alpha(t_j, Y(t_j)), \\ \bar{\beta}(s, Y) &= \beta(t_j, Y(t_j)). \end{aligned}$$

Let u satisfy the equation

$$\begin{aligned} u_t + \alpha u_x + \frac{\beta^2}{2} u_{xx} &= 0, \quad t < T, \\ u(x, T) &= g(x). \end{aligned} \quad (7)$$

The assumptions leading up to equation (6) imply that u and its derivatives exist. The Feynman-Kač formula shows that

$$u(x, t) = \mathbb{E}[g(X_T) | X_t = x]$$

and further

$$u(0, X_0) = \mathbb{E}[g(X_T)].$$

By the Itô formula

$$du(t, Y_t) = \left(u_t + \bar{\alpha} u_x + \frac{\bar{\beta}^2}{2} u_{xx} \right) (t, Y_t) dt + \bar{\beta} u_x(t, Y_t) dW,$$

and using equation (7)

$$\begin{aligned} du(t, Y_t) &= \left(-\alpha u_x - \frac{b^2}{2} u_{xx} + \bar{\alpha} u_x + \frac{\bar{b}^2}{2} u_{xx} \right) (t, Y_t) dt + \bar{\beta} u_x(t, Y_t) dW \\ &= \left((\bar{\alpha} - \alpha) u_x(t, Y_t) + \left(\frac{\bar{b}^2}{2} - \frac{b^2}{2} \right) u_{xx}(t, Y_t) \right) dt + \bar{\beta}(t, Y_t) u_x(t, Y_t) dW. \end{aligned}$$

We may now evaluate the integral from 0 to T as follows

$$u(T, Y_T) - u(0, X_0) = \int_0^T (\bar{\alpha} - \alpha) u_x(t, Y_t) dt + \int_0^T \frac{\bar{b}^2 - b^2}{2} u_{xx}(t, Y_t) dt + \int_0^T \bar{\beta}(t, Y_t) u_x(t, Y_t) dW.$$

Taking the expected value and using that $u(0, X_0) = \mathbb{E}[g(X_T)]$, we obtain

$$\begin{aligned} \mathbb{E} [g(Y_T) - g(X_T)] &= \int_0^T \mathbb{E} [(\bar{\alpha} - \alpha) u_x] + \frac{1}{2} \mathbb{E} [(\bar{\beta}^2 - \beta^2) u_{xx}] dt + \mathbb{E} \left[\int_0^T \bar{\beta} u_x dW \right] \\ &= \int_0^T \mathbb{E} [(\bar{\alpha} - \alpha) u_x] + \frac{1}{2} \mathbb{E} [(\bar{\beta}^2 - \beta^2) u_{xx}] dt. \end{aligned}$$

Since $\bar{\alpha}(t, Y) = \alpha(t_j, Y_{t_j})$, we have

$$f_1(t_j) = \mathbb{E}[(\bar{\alpha}(t_j, Y) - \alpha(t_j, Y_{t_j}))u_x(t_n, Y_{t_n})] = 0. \quad (8)$$

Let $a(t, x) = -(\alpha(t, x) - \alpha(t_j, Y_{t_j}))u_x(t, x)$, so that $f(t) = \mathbb{E}[a(t, Y_t)]$. Then by Itô's formula

$$\begin{aligned} \frac{\partial f}{\partial t} &= \frac{\partial}{\partial t} \mathbb{E}[a(t, Y_t)] \\ &= \mathbb{E}[\partial a(t, Y_t)] / \partial t \\ &= \mathbb{E} \left[\left(a_t + \bar{\alpha} a_x + \frac{\bar{\beta}^2}{2} a_{xx} \right) dt + a_x \bar{\beta} dW \right] / \partial t \\ &= \mathbb{E} \left[a_t + \bar{\alpha} a_x + \frac{\bar{\beta}^2}{2} a_{xx} \right] \\ &= \mathcal{O}(1). \end{aligned}$$

Therefore, there exists a constant $C \in \mathbb{R}$ such that $|f_1'(t)| \leq C$ for $t_j \leq t \leq t_n + 1$. Together with the initial condition in equation (8), this implies that

$$f_1(t) \equiv \mathbb{E}[(\bar{\alpha}(t, Y) - \alpha(t, Y_t))u_x(t, Y_t)] = \mathcal{O}(\Delta t_j),$$

for $t_j \leq t \leq t_{j+1}$. Similarly for f_2 we get

$$f_2(t) \equiv \mathbb{E}[(\bar{\beta}^2(t, Y) - \beta^2(t, Y_t))u_{xx}(t, Y_t)] = \mathcal{O}(\Delta t_j).$$

Thus the order of convergence is in this case $\mathcal{O}(\Delta t) = \mathcal{O}(h)$.

3 Change of Measure

In financial mathematics it is useful to be able to change from the physical measure to the risk-neutral measure. In the method of importance sampling, one changes probability measure when going from the original distribution function to the importance sampling distribution function. The Girsanov theorem describes how the dynamics of stochastic processes change when going from one probability measure to another.

Theorem 3.1 (Girsanov Theorem). *Let b be an $\mathbb{R}$ -valued process adapted to $\{\mathcal{F}_t^W\}$ satisfying*

$$\int_0^t \|b(s)\|^2 ds < \infty$$

for $t \in [0, T]$, and let

$$X(t) = \exp \left(-\frac{1}{2} \int_0^t \|b(s)\|^2 ds + \int_0^t b(s) dW(s) \right).$$

If $E_P[X(T)] = 1$, then $\{X(t), t \in [0, T]\}$ is a martingale and the measure Q on $(\Omega, \mathcal{F}_T^W)$ defined by

$$\frac{dQ}{dP} = X(T)$$

is equivalent to P . Under Q , the process

$$W^Q(t) := W(t) - \int_0^t b(s) ds, \quad t \in [0, T]$$

is a standard Brownian motion with respect to the filtration $\{\mathcal{F}_t^W\}$.

For a reference, see for example Glasserman (2003).

Example 3.1. Let

$$X_t = -\mu t + W_t, \quad \mu > 0,$$

where W is a Brownian motion. By using the Girsanov theorem, we want to show that the distribution of the random variable

$$M := \sup_{0 \leq t < \infty} X_t$$

is exponentially distributed.

For $b > 0$, define $\tau_b = \inf\{t \geq 0 : W_t \geq b\}$ as the first passage time for a Brownian motion over b . Let $\eta_t := \exp\left\{-\mu W_t - \frac{\mu^2}{2}t\right\}$ and let $\tilde{P}$ be a measure that satisfies

$$\tilde{P}(B) = E[\mathbb{1}_B \eta_t]$$

for $B \in \mathcal{F}_t$. Further, let $\tilde{W}_t := \mu t + W_t$. By the Girsanov theorem, $W_t = \tilde{W}_t - \mu t$ is a Brownian motion with drift $-\mu$ under $\tilde{P}$ and

$$\tilde{P}(\tau_b \leq t) = E[\mathbb{1}_{\tau_b \leq t} \eta_t].$$

On the set $\{\tau_b \leq t\} \in \mathcal{F}_t^W \cap \mathcal{F}_{\tau_b}^W = \mathcal{F}_{t \wedge \tau_b}^W$ we have $\eta_{t \wedge \tau_b} = \eta_{\tau_b}$. We can deduce that

$$\begin{aligned}
\tilde{P}(\tau_b \leq t) &= \mathbb{E}[\mathbb{1}_{\tau_b \leq t} \eta_t] = \mathbb{E}[\mathbb{1}_{\tau_b \leq t} \mathbb{E}[\eta_t | \mathcal{F}_{t \wedge \tau_b}^W]] \\
&= \mathbb{E}[\mathbb{1}_{\tau_b \leq t} \eta_{t \wedge \tau_b}] = \mathbb{E}[\mathbb{1}_{\tau_b \leq t} \eta_{\tau_b}] \\
&= \mathbb{E}\left[\mathbb{1}_{\tau_b \leq t} \exp\left\{-\mu b - \frac{1}{2}\mu^2 \tau_b\right\}\right] \\
&= \int_0^t \exp\left\{-\mu b - \frac{\mu^2 s}{2}\right\} P(\tau_b \in ds),
\end{aligned} \tag{9}$$

and in the same way

$$\tilde{P}(\tau_b < \infty) = \int_0^\infty \exp\left\{-\mu b - \frac{\mu^2 s}{2}\right\} P(\tau_b \in ds). \tag{10}$$

We need to show that

$$P(\tau_b \leq t) = P(\sup_{0 \leq s \leq t} W_s \geq b) = 2P(W_t \geq b).$$

According to the reflection principle, see Chapter 2.6.A in (Karatzas & Shreve, 1998), we have

$$P(\tau_b \leq t) = P(\tau_b \leq t, W_t > b) + P(\tau_b \leq t, W_t < b).$$

If $W_t > b$, then also $\tau_b \leq t$. Thus

$$P(\tau_b \leq t, W_t > b) = P(W_t > b).$$

On the other hand, if $W_t > b$ and $\tau_b \leq t$, then the process has reached level b sometime before time t and then traveled to some point below b , call this point c . Because of symmetry, the probability of doing this is the same as the probability of W_t going from b to the point $2b - c$. Hence,

$$P(\tau_b \leq t, W_t > b) = P(\tau_b \leq t, W_t < b) = P(W_t > b)$$

and thus

$$P(\tau_b \leq t) = P(\tau_b \leq t, W_t > b) + P(\tau_b \leq t, W_t < b) = 2P(W_t > b).$$

We know that

$$2P(W_t > b) = \sqrt{\frac{2}{\pi}} \int_{b/\sqrt{t}}^\infty e^{-x^2/2} dx.$$

Differentiating with respect to t gives us the density of the passage time

$$P(\tau_b \in dt) = \frac{b}{\sqrt{2\pi t^3}} \exp\left\{-\frac{b^2}{2t}\right\} dt, \quad t > 0 \tag{11}$$

and

$$\mathbb{E}[e^{-\alpha \tau_b}] = e^{-b\sqrt{2\alpha}}. \tag{12}$$

To see this, let

$$u(x) = E_x[e^{-\alpha \tau_b}].$$

Then $u(x)$ is the solution to the ODE

$$\begin{aligned}\frac{1}{2}u_{xx} - \alpha u &= 0, \\ u(-\infty) &= 0, \\ u(b) &= 1.\end{aligned}$$

Thus, using that $x = 0$, we get

$$u(x) = Ce^{\sqrt{2\alpha}x} + De^{-\sqrt{2\alpha}x} = e^{\sqrt{2\alpha}(x-b)} = e^{-b\sqrt{2\alpha}}.$$

Using equation (9) and (11),

$$\tilde{P}(\tau_b \in dt) = \frac{b}{\sqrt{2\pi t^3}} \exp\left\{-\frac{(b - \mu t)^2}{2t}\right\} dt, \quad t > 0.$$

Equation (10) implies that

$$\tilde{P}(\tau_b < \infty) = e^{\alpha b} \mathbb{E} \left[\exp\left\{-\frac{1}{2}\mu^2 \tau_b\right\} \right]$$

and using equation (12), we get

$$\tilde{P}(\tau_b < \infty) = e^{-2\mu b}.$$

Thus,

$$\tilde{P}\left(\sup_{0 \leq t < \infty} W_t \geq b\right) = \tilde{P}(\tau_b < \infty) = e^{-2\mu b}.$$

Since this probability is of a Brownian motion under the measure $\tilde{P}$, the probability of a Brownian motion with drift under the original probability measure P is also

$$P(M \geq b) = P\left(\sup_{0 \leq t < \infty} -\mu t + W_t \geq b\right) = e^{-2\mu b}.$$

4 Stochastic Optimal Control Problem

Stochastic optimal control models deal with the problem of finding a control law for a given system such that a certain optimality criterion is achieved. We have a state process X and optimal control processes b with certain control constraints. Given a controlled stochastic differential equation, each choice of the control parameter yields a different stochastic variable as a solution to the stochastic differential equation. Each pathwise trajectory of this stochastic process has an associated cost, and we seek to minimize the expected cost over all choices of the control parameter (Kafash & Nadizadeh, 2017).

The Hamilton-Jacobi-Bellman equation, often referred to as the HJB equation, is a sufficient condition for the optimal control problem. This result is concluded in the verification theorem for dynamic programming. Our presentation follows the same lines as in Björk (2009).

Let $\mu(t, x, b)$ and $\sigma(t, x, b)$ be given functions of the form

$$\begin{aligned}\mu &: \mathbb{R}_+ \times \mathbb{R}^n \times \mathbb{R}^k \rightarrow \mathbb{R}^n, \\ \sigma &: \mathbb{R}_+ \times \mathbb{R}^n \times \mathbb{R}^k \rightarrow \mathbb{R}^{n \times d}.\end{aligned}$$

Consider the following controlled stochastic differential equation

$$\begin{aligned}dX_t &= \mu(t, X_t, b_t)dt + \sigma(t, X_t, b_t)dW_t, \\ X_0 &= x_0\end{aligned}$$

for a given point $x_0 \in \mathbb{R}^n$ for the n -dimensional state process X . Here, W is a d -dimensional Brownian motion and we try to control the state process with the control process $b \in \mathbb{R}^k$.

In the following theorem and proof, $\mathcal{A}^b$ denotes the partial differential operator defined by

$$\mathcal{A}^b = \sum_{i=1}^n \mu_i^b(t, x) \frac{\partial}{\partial x_i} + \frac{1}{2} \sum_{i,j=1}^n C_{i,j}^b(t, x) \frac{\partial^2}{\partial x_i \partial x_j}$$

for any fixed vector b , where $\mu^b(t, x) = \mu(t, x, b)$ and $C^b(t, x) = \sigma(t, x, b)\sigma(t, x, b)'$. Here, $'$ denotes the matrix transpose.

In most cases it is natural to require that the control process b is adapted to the process X . One may choose a deterministic function $g(t, x)$

$$g : \mathbb{R}_+ \times \mathbb{R}^n \rightarrow \mathbb{R}^k$$

to obtain an adapted control process and then define the control process b by

$$b_t = g(t, X_t).$$

We restrict ourselves to such control laws. In this section, we will from now on use boldface to indicate that $\mathbf{b}$ is a function, and italics to denote the value b of a control at a certain time.

We also want to satisfy some control constraints and thus we take a given subset $B \subset \mathbb{R}^k$ and require that $b_t \in B$ for each t . We denote the class of admissible control laws by $\mathcal{B}$. We say that a control law $\mathbf{b}$ is admissible if $\mathbf{b}(t, x) \in B$ for all $t \in \mathbb{R}_+$ and all $x \in \mathbb{R}^n$, and for any given initial point (t, x) the stochastic differential equation

$$\begin{aligned} dX_s &= \mu(s, X_s, \mathbf{b}(s, X_s))dt + \sigma(s, X_s, \mathbf{b}(s, X_s))dW_s, \\ X_t &= x \end{aligned}$$

has a unique solution.

Consider a given pair of functions

$$\begin{aligned} F &: \mathbb{R}_+ \times \mathbb{R}^n \times \mathbb{R}^k \rightarrow \mathbb{R}, \\ \Phi &: \mathbb{R}^n \rightarrow \mathbb{R}. \end{aligned}$$

Let the value function

$$\mathcal{J} : \mathbb{R}_+ \times \mathbb{R}^n \times \mathcal{B} \rightarrow \mathbb{R}$$

be defined by

$$\mathcal{J}(t, x, \mathbf{b}) := \mathbb{E} \left[\int_t^T F(s, X_s^{\mathbf{b}}, \mathbf{b}_s) ds + \Phi(X_T^{\mathbf{b}}) \right]$$

given the dynamics

$$\begin{aligned} dX_s^{\mathbf{b}} &= \mu(s, X_s^{\mathbf{b}}, \mathbf{b}(s, X_s^{\mathbf{b}}))ds + \sigma(s, X_s^{\mathbf{b}}, \mathbf{b}(s, X_s^{\mathbf{b}}))dW_s, \\ X_t &= x. \end{aligned}$$

The formal problem is to maximize the value function over all $\mathbf{b} \in \mathcal{B}$. Thus the optimal value function

$$\hat{\mathcal{J}} : \mathbb{R}_+ \times \mathbb{R}^n \rightarrow \mathbb{R}$$

is defined by

$$\hat{\mathcal{J}}(t, x) := \sup_{\mathbf{b} \in \mathcal{B}} \mathcal{J}(t, x, \mathbf{b}).$$

If there exist an admissible control law $\hat{\mathbf{b}}$ such that

$$\mathcal{J}(t, x, \hat{\mathbf{b}}) = \hat{\mathcal{J}}(t, x),$$

then $\hat{\mathbf{b}}$ is an optimal control law for the given problem. You may read more about stochastic optimal control in Björk (2009). We will now state the theorem.

Theorem 4.1 (Verification Theorem for the Hamilton-Jacobi-Bellman equation). *Suppose that we have a sufficiently integrable function $u(t, x)$ solving the HJB equation*

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) + \sup_{b \in B} \{F(t, x, b) + \mathcal{A}^b u(t, x)\} = 0, & \forall (t, x) \in (0, T) \times \mathbb{R}^n \\ u(t, x) = \Phi(x), & \forall x \in \mathbb{R}^n \end{cases}$$

and an admissible control law function $g(t, x)$ such that for each fixed (t, x) , the supremum

$$\sup_{b \in B} \{F(t, x, b) + \mathcal{A}^b u(t, x)\}$$

is attained by choosing $b = g(t, x)$. Then the optimal value function $\hat{\mathcal{J}}$ to the control problem is given by

$$\hat{\mathcal{J}}(t, x) = u(t, x)$$

and there exists an optimal control law $\hat{\mathbf{b}}(t, x) = g(t, x)$. Here, B denotes the set of control constraints which we allow to be state and time dependent, i.e., of the form $B(t, x)$.

Proof. Choose an arbitrary control law $\mathbf{b} \in \mathcal{B}$ from the set of admissible controls, and a fix a point (t, x) . Define the process $X^{\mathbf{b}}$ on the interval $[t, T]$ as the solution to the equation

$$\begin{aligned} dX_s^{\mathbf{b}} &= \mu^{\mathbf{b}}(s, X_s^{\mathbf{b}})ds + \sigma^{\mathbf{b}}(s, X_s^{\mathbf{b}})dW_s, \\ X_t &= x. \end{aligned}$$

Assuming the functions u and g are as in Theorem 4.1, insert the process $X^{\mathbf{b}}$ into u and use the Itô formula to obtain

$$\begin{aligned} u(T, X_T^{\mathbf{b}}) &= u(t, x) + \int_t^T \left(\frac{\partial u}{\partial t}(s, X_s^{\mathbf{b}}) + (\mathcal{A}^{\mathbf{b}}u)(s, X_s^{\mathbf{b}}) \right) ds \\ &\quad + \int_t^T \nabla_x u(s, X_s^{\mathbf{b}}) \sigma^{\mathbf{b}}(s, X_s^{\mathbf{b}}) dW_s. \end{aligned}$$

Since u solves the HJB equation, we have for all $b \in B$

$$\frac{\partial u}{\partial t}(t, x) + F(t, x, b) + \mathcal{A}^b u(t, x) \leq 0$$

implying

$$\frac{\partial u}{\partial t}(s, X_s^{\mathbf{b}}) + (\mathcal{A}^{\mathbf{b}}u)(s, X_s^{\mathbf{b}}) \leq -F^{\mathbf{b}}(s, X_s^{\mathbf{b}})$$

for each s almost surely with respect to the probability measure. The boundary condition $u(t, x) = \Phi(x)$ implies that $u(T, X_T^{\mathbf{b}}) = \Phi(X_T^{\mathbf{b}})$, and thus

$$u(t, x) \geq \int_t^T F^{\mathbf{b}}(s, X_s^{\mathbf{b}})ds + \Phi(X_T^{\mathbf{b}}) - \int_t^T \nabla_x u(s, X_s^{\mathbf{b}}) \sigma^{\mathbf{b}} dW_s.$$

Assuming enough integrability, the stochastic integral vanishes and after taking expectations we are left with

$$u(t, x) \geq \mathbb{E}_{t,x} \left[\int_t^T F^{\mathbf{b}}(s, X_s^{\mathbf{b}})ds + \Phi(X_T^{\mathbf{b}}) \right] = \mathcal{J}(t, x, \mathbf{b}),$$

with $\mathcal{J}$ denoting the value function. Since $\mathbf{b}$ is arbitrary we get

$$u(t, x) \geq \sup_{\mathbf{b} \in \mathcal{B}} \mathcal{J}(t, x, \mathbf{b}) = \hat{\mathcal{J}}(t, x). \quad (13)$$

To show the reverse inequality, we choose $\mathbf{b}(t, x) = \mathbf{g}(t, x)$. By assumption

$$\frac{\partial u}{\partial t}(t, x) + F^{\mathbf{g}}(t, x) + \mathcal{A}^{\mathbf{g}}u(t, x) = 0,$$

and after doing similar calculations as before we get

$$u(t, x) = \mathbb{E}_{t,x} \left[\int_t^T F^{\mathbf{g}}(s, X_s^{\mathbf{g}}) ds + \Phi(X_T^{\mathbf{g}}) \right] = \mathcal{J}(t, x, \mathbf{g}). \quad (14)$$

From equation (13) we have

$$u(t, x) \geq \hat{\mathcal{J}}(t, x)$$

and since $\hat{\mathcal{J}}(t, x)$ is the optimal value function, we have

$$\hat{\mathcal{J}}(t, x) \geq \mathcal{J}(t, x, \mathbf{g}).$$

These two inequalities and equation (14) prove that $u(t, x) = \hat{\mathcal{J}}(t, x)$ and $\mathbf{g}$ is the optimal control law. $\square$

Remark. Instead of a maximization problem, one might consider a minimization problem. With the value function and optimal value function adjusted accordingly, it is easy to see that the above results still hold if the expression

$$\sup_{b \in B} \{F(t, x, b) + \mathcal{A}^b u(t, x)\}$$

in the Hamilton-Jacobi-Bellman equation is replaced by the expression

$$\inf_{b \in B} \{F(t, x, b) + \mathcal{A}^b u(t, x)\}.$$

5 Importance Sampling for Diffusions

5.1 Introductory Problem

Let W be a Brownian motion and let $dX = \mu(t, X_t)dt + \sigma(t, X_t)dW$. Consider the problem of calculating the probability that the process X is larger than a certain non-negative barrier B at time T

$$P_x(X_T \geq B) := p.$$

Such a probability can be calculated efficiently using partial differential equation methods. In fact $p = u(0, x)$, where $u(t, x)$ solves

$$\begin{cases} \frac{\partial u}{\partial t} + \mu \frac{\partial u}{\partial x} + \frac{\sigma^2}{2} \frac{\partial^2 u}{\partial x^2} = 0, \\ u(T, x) = \Psi(x) \end{cases}$$

with

$$\Psi(x) = \begin{cases} 1, & x \geq B, \\ 0, & x < B. \end{cases}$$

Nevertheless, we keep this relatively easy problem as a model problem and remark that the current set-up can easily be generalized to more complicated settings involving higher-dimensional diffusions as well as path-dependent features.

Define the indicator function $\mathbb{1}_A : X \rightarrow \{0, 1\}$ of a subset A of a set X as

$$\mathbb{1}_A(x) := \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Continuing on the introductory Section 2.2 about importance sampling, the Monte Carlo estimation $\hat{p}$ of p would be

$$\hat{p} = \frac{1}{N} \sum_{k=1}^N I_k$$

where $I_1, \dots, I_N$ are independent observations of $\mathbb{1}_{X_T \geq B}$. Note that $E[\hat{p}] = p$. We have further interest in the variance of this probability, which we want to minimize by importance sampling. Let

$$dW^b = dW - b(t, X_t)dt$$

for some non-negative function $b(t, X_t)$, so that

$$dX = \mu(t, X_t)dt + \sigma(t, X_t)(dW^b + b(t, X_t)dt).$$

We perform a change of measure according to the Girsanov theorem as follows

$$\frac{dP^b}{dP} = \exp \left\{ -\frac{1}{2} \int_0^T b^2 dt + \int_0^T b dW \right\} = \exp \left\{ \frac{1}{2} \int_0^T b^2 dt + \int_0^T b dW^b \right\}.$$

We get

$$\mathbb{E}[\hat{p}] = P(X_T \geq B) = \mathbb{E}[\mathbb{1}_{X_T \geq B}] = \mathbb{E}^b \left[\mathbb{1}_{\{X_T \geq B\}} \frac{dP}{dP^b} \right] = \mathbb{E}^b[\hat{p}^b].$$

Now

$$dX = (\mu(t, X_t) + b(t, X_t)\sigma(t, X_t)) dt + \sigma(t, X_t)dW^b$$

where dW^b is a Brownian motion under the b -measure. The approximated probability under the new measure is

$$\hat{p}^b = \frac{1}{N} \sum_{k=1}^N I_k^b \frac{dP}{dP^b}$$

where $I_1^b, \dots, I_N^b$ are independent observations of $\mathbb{1}_{X_T \geq B}^b$. We wish to see that

$$\text{Var}^b[\hat{p}^b] \leq \text{Var}[\hat{p}].$$

Thus, we choose b to minimize the variance

$$\text{Var}^b \left[\mathbb{1}_{\{X_T \geq B\}} \frac{dP}{dP^b} \right] = \mathbb{E}^b \left[\left(\mathbb{1}_{\{X_T \geq B\}} \frac{dP}{dP^b} \right)^2 \right] - \left(\mathbb{E}^b \left[\mathbb{1}_{\{X_T \geq B\}} \frac{dP}{dP^b} \right] \right)^2.$$

We know that

$$\left(\mathbb{E}^b \left[\mathbb{1}_{\{X_T \geq B\}} \frac{dP}{dP^b} \right] \right)^2 = p^2$$

and thus only need to consider the second moment

$$\mathbb{E}^b \left[\left(\mathbb{1}_{\{X_T \geq B\}} \frac{dP}{dP^b} \right)^2 \right] = \mathbb{E}^b \left[\mathbb{1}_{\{X_T \geq B\}} \exp \left\{ -\int_0^T b^2 dt - 2 \int_0^T b dW^b \right\} \right]. \quad (15)$$

Let us perform a new change of measure to simplify the expression of the second moment. We have

$$\frac{dQ}{dP^b} = \exp \left\{ -2 \int_0^T b^2 dt - 2 \int_0^T b dW^b \right\}$$

with

$$dW^Q = dW^b + 2b(t, X_t)dt$$

which now is a Brownian motion under Q , and

$$dX = \mu(t, X_t)dt + \sigma(t, X_t)(dW^Q - b(t, X_t)dt).$$

The second moment in equation (15) then becomes

$$\mathbb{E}^Q \left[\mathbb{1}_{\{X_T \geq B\}} e^{\int_0^T b^2 dt} \right].$$

5.1.1 Constant Coefficients

Let μ, b and σ be constant. In this case, $p = P_x(X_T \geq B)$ can actually be expressed in terms of the cumulative distribution function for normal distribution, but we continue on this example to get an overview of the behaviour of the variance after applying importance sampling. We have

$$dX = \mu dt + \sigma(dW^Q - bdt) = (\mu - \sigma b)dt + \sigma dW^Q.$$

Again, we look at minimizing the second moment

$$\mathbb{E}^Q \left[\mathbb{1}_{\{X_T \geq B\}} e^{\int_0^T b^2 dt} \right] =: f(b),$$

and we get

$$\inf_b \mathbb{E}^Q \left[\mathbb{1}_{\{X_T \geq B\}} e^{b^2 T} \right] = \inf_b e^{b^2 T} Q(X_T \geq B).$$

We know that

$$f(b) \geq p^2.$$

To simplify, let $\sigma = 1$. Then we want to minimize

$$\begin{aligned} f(b) &= e^{b^2 T} Q(X_T \geq B) = e^{b^2 T} Q((\mu - b)T + dW_T^Q \geq B) \\ &= e^{b^2 T} Q(dW_T^Q \geq B - \mu T + bT) = e^{b^2 T} \int_{B - \mu T + bT}^{\infty} \varphi(y) dy. \end{aligned}$$

What is interesting is to analyze whether $f(b)$ has a minimum, which would imply that the change of measure is optimal in the class of constant drifts.

To get an overview of the problem, let us look at the simple example when $T = 1$ and $\mu = 0$. Taking the derivative of

$$f(b) = e^{b^2} \int_{B+b}^{\infty} \varphi(y) dy$$

we get

$$\begin{aligned} f'(b) &= 2be^{b^2} \int_{B+b}^{\infty} \varphi(y) dy - e^{b^2} \varphi(B+b) \\ &= e^{b^2} (2b \Phi(-(B+b)) - \varphi(B+b)), \end{aligned}$$

and setting $f'(b) = 0$ to find a minimum, we simplify to

$$h(b) := 2b \Phi(-(B+b)) - \varphi(B+b) = 0.$$

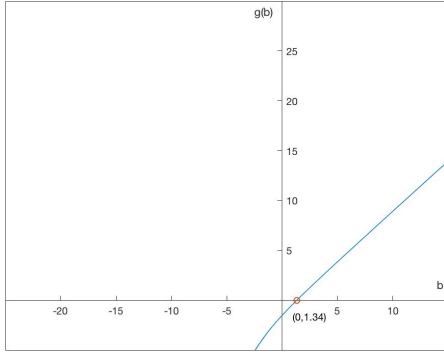
We will not be able to analytically find an explicit solution to $h(b) = 0$, but we expect that there will be a root. Let

$$g(b) := \frac{h(b)}{\Phi(-(B+b))} = 2b - \frac{\varphi(B+b)}{\Phi(-(B+b))}.$$

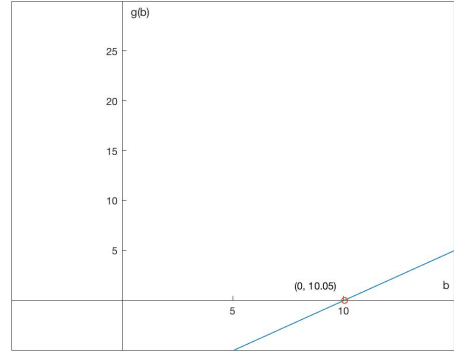
Then

$$\begin{aligned}
g'(b) &= 2 - \frac{-(B+b)\varphi(B+b)\Phi(-(B+b)) - \varphi(B+b)\varphi(-(B+b))}{(\Phi(-(B+b)))^2} \\
&= 2 + \frac{(B+b)\varphi(B+b)\Phi(-(B+b)) + \varphi(B+b)\varphi(-(B+b))}{(\Phi(-(B+b)))^2} \geq 0.
\end{aligned}$$

Note that both $B > 0$ and $b > 0$, this and the fact that the normal cumulative distribution function as well as the probability density function is non-negative implies the inequality. Thus, $g(b)$ is a monotone function and therefore has at most one root. Let b_{opt} be such that $g(b_{\text{opt}}) = 0$, then also $h(b_{\text{opt}}) = 0$. Thus, b_{opt} optimizes f . In Figure 4, the function g is plotted and numerical values of b_{opt} are determined for two different values of B .



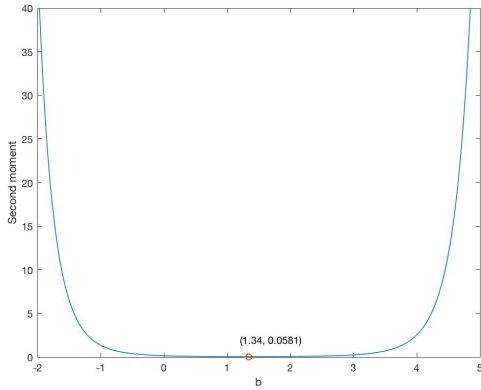
(a) $B = 1, \mu = 0, b_{\text{opt}} \approx 1.34$



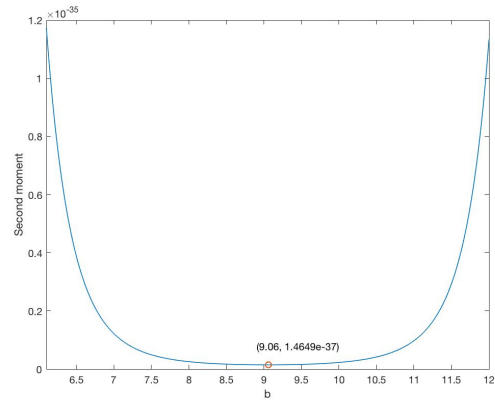
(b) $B = 10, \mu = 0, b_{\text{opt}} \approx 10.05$

Figure 4: Modified version g of the second moment f to find optimal b

In Figure 5, the second moment depending on the new added drift b can be seen. Still, $\sigma = 1$ and $T = 1$.



(a) $B = 1, \mu = 0, b_{\text{opt}} \approx 1.34$



(b) $B = 10, \mu = 1, b_{\text{opt}} \approx 9.06$

Figure 5: Second moment depending on b

Remark. It seems that the optimum drift $b_{\text{opt}} \approx \frac{B}{T}$, which can be interpreted as half of the simulated paths being below and the other half above the barrier at time T .

Remark. In Figure 5 (b), if we would instead have $\mu = 0$ we would get $b_{\text{opt}} \approx 10.05$ as in Figure 4 (b).

5.2 Stochastic Process with Controlled Drift

Now we consider a more involved example exhibiting path dependencies. To some extent, however, this actually simplifies the problem since it no longer depends on a finite horizon. We begin with a Brownian motion with drift

$$dX_t = \mu dt + \sigma dW_t$$

and we are interested in the probability

$$p(x) := P_x \left(\inf_{s \geq 0} X_s \leq 0 \right). \quad (16)$$

Going back to Example 3.1 which is similar to the problem of estimating this probability, we already have a solution, i.e., $p(x)$ is known. In this set up we have a stochastic process starting at $X_0 = x \geq 0$ where we are interested in the probability of the infimum of the process X_t being below zero. In Example 3.1, we instead start at $X_0 = x = 0$ and calculate the probability of hitting a barrier. Due to the reflection principle, these probabilities will be the same. Nevertheless, we keep this relatively easy problem as a model problem and remark that the current set-up can easily be generalized to more complicated settings.

With a problem set up inspired by Jeanblanc-Picqué and Shiryaev (1995), we perform a change of measure, with $X = (X_t)_{t \geq 0}$, and let

$$dX_t = \mu dt + \sigma dW_t = \mu dt + \sigma dW_t^b - dZ_t$$

where $W = (W_t)_{t \geq 0}$ is a standard Brownian motion. Also, $Z = (Z_t)_{t \geq 0}$ is a non-negative, non-decreasing, adapted process such that

$$dZ_t = b(X_t)dt, \quad Z_0 = Z_0(x),$$

where $b = b(x)$, $Z_0 = Z_0(x)$ are arbitrary measurable functions satisfying $0 \leq b(x) \leq K < \infty$, $0 \leq Z_0(x) \leq x$. Assuming $X_0 = x \geq 0$, we wonder when the process X hits zero, call this moment τ . For $t \geq \tau$, let $dX_t = dZ_t = 0$.

After the previously mentioned change of measure, and for simplicity letting $\sigma = 1$, we get

$$\begin{aligned} dX_t &= \mu dt + dW_t^b - dZ_t = \mu dt + dW_t^b - b(X_t)dt \\ &= (\mu - b(X_t))dt + dW_t^b \end{aligned}$$

with

$$\begin{aligned} dW_t^b &= dX - (\mu - b(X_t))dt = \mu dt + dW_t - (\mu - b(X_t))dt \\ &= b(X_t)dt + dW. \end{aligned}$$

Then

$$\mathbb{E} [\mathbb{1}_{\tau_0 < \infty}] = \mathbb{E}^b \left[\mathbb{1}_{\tau_0 < \infty} \frac{dP}{dP^b} \right]$$

where

$$\begin{aligned} \frac{dP^b}{dP} &= e^{-\frac{1}{2} \int_0^{\tau_0} b^2 dt - \int_0^{\tau_0} b dW} = e^{-\frac{1}{2} \int_0^{\tau_0} b^2 dt - \int_0^{\tau_0} u(dW^b - bdt)} \\ &= e^{\frac{1}{2} \int_0^{\tau_0} b^2 dt - \int_0^{\tau_0} b dW^b}. \end{aligned}$$

Again, we have the variance

$$\begin{aligned} \text{Var}^b \left[\mathbb{1}_{\tau_0 < \infty} \frac{dP}{dP^b} \right] &= \mathbb{E}^b \left[\left(\mathbb{1}_{\tau_0 < \infty} \frac{dP}{dP^b} \right)^2 \right] - \left(\mathbb{E}^b \left[\mathbb{1}_{\tau_0 < \infty} \frac{dP}{dP^b} \right] \right)^2 \\ &= \mathbb{E}^b \left[\left(\mathbb{1}_{\tau_0 < \infty} \frac{dP}{dP^b} \right)^2 \right] - p^2 \end{aligned}$$

and we are interested in the second moment

$$\begin{aligned} \mathbb{E}^b \left[\left(\mathbb{1}_{\tau_0 < \infty} \frac{dP}{dP^b} \right)^2 \right] &= \mathbb{E}^b \left[\left(\mathbb{1}_{\tau_0 < \infty} e^{-\frac{1}{2} \int_0^{\tau_0} b^2 dt + \int_0^{\tau_0} b dW^b} \right)^2 \right] \\ &= \mathbb{E}^b \left[\mathbb{1}_{\tau_0 < \infty} e^{-\int_0^{\tau_0} b^2 dt + 2 \int_0^{\tau_0} b dW^b} \right]. \end{aligned} \tag{17}$$

We perform a new change of measure

$$\frac{dQ}{dP^b} = e^{-2 \int_0^{\tau_0} b^2 dt + 2 \int_0^{\tau_0} b dW^b}$$

with

$$dW^Q = -2bdt + dW^b$$

and

$$dX_t = \mu dt - bdt + dW_t^b = \mu dt - bdt + dW_t^Q + 2bdt = \mu dt + bdt + dW_t^Q.$$

The second moment from equation (17) becomes

$$\begin{aligned} \mathbb{E}^b \left[\mathbb{1}_{\tau_0 < \infty} e^{-\int_0^{\tau_0} b^2 dt + 2 \int_0^{\tau_0} b dW^b} \right] &= \mathbb{E}^Q \left[\mathbb{1}_{\tau_0 < \infty} e^{-\int_0^{\tau_0} b^2 dt + 2 \int_0^{\tau_0} b dW^b} \frac{dP^b}{dQ} \right] \\ &= \mathbb{E}^Q \left[\mathbb{1}_{\tau_0 < \infty} e^{\int_0^{\tau_0} b^2 dt} \right]. \end{aligned}$$

We want to minimize the variance of the expected value of the probability that the infimum of X_t is less than 0. Again, we only need to minimize the second moment

$$\mathbb{E}^Q \left[\mathbb{1}_{\{\tau_0 < \infty\}} e^{\int_0^{\tau_0} b^2 dt} \right] := f(x) \geq p^2. \tag{18}$$

5.2.1 Constant Push in Specified Interval

Let us continue on the problem set up from previous section inspired by Jeanblanc-Picqué and Shiryaev (1995). Now consider a stochastic process with a controlled drift that pushes the diffusion process with a constant $K \in \mathbb{R}$ when x is less than a certain $\hat{x}$. We want to find the optimal "push" K given a fixed level $\hat{x}$ for

$$b(x) = \begin{cases} K, & x < \hat{x}, \\ 0, & x \geq \hat{x}. \end{cases}$$

Further, we may analyze how $f(x)$ from equation (18) behaves and if there is some constant K that minimizes the variance.

If $f(x)$ can be represented as in equation (18), then by the Feynman-Kač formula it is a solution to the following boundary value problem

$$\frac{1}{2}f_{xx} + (\mu - b(x))f_x + (b(x))^2f = 0$$

which results in a system of ordinary differential equations

$$\begin{cases} \frac{1}{2}f_{xx} + (\mu - K)f_x + K^2f = 0, & x < \hat{x} \\ \frac{1}{2}f_{xx} + \mu f_x = 0, & x \geq \hat{x}. \end{cases}$$

We solve for f when $x < \hat{x}$ and get a constant-coefficient homogeneous ordinary differential equation. Assume $f = e^{rx}$ is a solution, then

$$\frac{1}{2}r^2e^{rx} + (\mu - K)re^{rx} + K^2e^{rx} = 0$$

which simplifies to

$$r^2 + 2(\mu - K)r + 2K^2 = 0. \tag{19}$$

Thus,

$$r = K - \mu \pm \sqrt{\mu^2 - 2\mu K - K^2}$$

resulting in three different cases depending on if the characteristic polynomial in equation (19) has two distinct roots, a double root or two complex conjugate roots. We restrict K such that

$$\mu^2 - 2\mu K - K^2 > 0 \tag{20}$$

and get the solution

$$f = C_1e^{r_1x} + C_2e^{r_2x}, \quad x < \hat{x}$$

for two distinct $r_1, r_2 \in \mathbb{R}$.

Further, when $x \geq \hat{x}$, we have

$$\frac{1}{2}f_{xx} + \mu f_x = 0$$

which can be rearranged as

$$f_{xx} = -2\mu f_x$$

resulting in the first order derivative being

$$f_x = e^{-2\mu x}.$$

Hence,

$$f = C_3 e^{-2\mu x} + C_4, \quad x \geq \hat{x}.$$

The system becomes

$$f = \begin{cases} C_1 e^{r_1 x} + C_2 e^{r_2 x}, & x < \hat{x}, \\ C_3 e^{-2\mu x} + C_4, & x \geq \hat{x} \end{cases}$$

for some real-valued constants C_1, C_2, C_3 and C_4 . To solve the system of ordinary differential equations and find values for C_1, C_2, C_3 and C_4 we need at least four boundary conditions. We may use the initial value of f and how the second moment is expected to behave at infinity. Also, both the function f and its derivative f_x need to be continuous in the point $\hat{x}$. We get the following boundary conditions

$$\begin{aligned} f(0) &= 1, \\ f(\infty) &= 0, \\ C_1 e^{r_1 \hat{x}} + C_2 e^{r_2 \hat{x}} &= C_3 e^{-2\mu \hat{x}} + C_4, \\ C_1 r_1 e^{r_1 \hat{x}} + C_2 r_2 e^{r_2 \hat{x}} &= -C_3 2\mu e^{-2\mu \hat{x}}. \end{aligned}$$

The second condition implies that $C_4 = 0$. Further, from condition one, three and four we get

$$\begin{aligned} C_1 + C_2 &= 1, \\ C_1 e^{r_1} + C_2 e^{r_2} &= C_3 e^{-2\mu}, \\ C_1 r_1 e^{r_1} + C_2 r_2 e^{r_2} &= -C_3 2\mu e^{-2\mu} \end{aligned}$$

resulting in

$$\begin{cases} C_1 = \frac{-e^{r_2(2\mu+r_2)}}{e^{r_1(2\mu+r_1)} - e^{r_2(2\mu+r_2)}} \\ C_2 = 1 - C_1 \\ C_3 = (C_1 e^{r_1} + C_2 e^{r_2}) e^{2\mu} \\ C_4 = 0. \end{cases}$$

So far, this is a general example. Let $K = 3$, $\mu = 10$ and $\hat{x} = 1$. The resulting function f can be seen in Figure 6.

Note that K only affects f for $x < \hat{x}$. To optimize the second moment, and hence minimize the variance, we need f to be small. When x is close to 0, we therefore want to "push" the stochastic process towards 0. Thus, we expect the optimal K to be the smallest possible K such that equation (20) still holds, i.e., the largest possible negative drift with regard to absolute value.

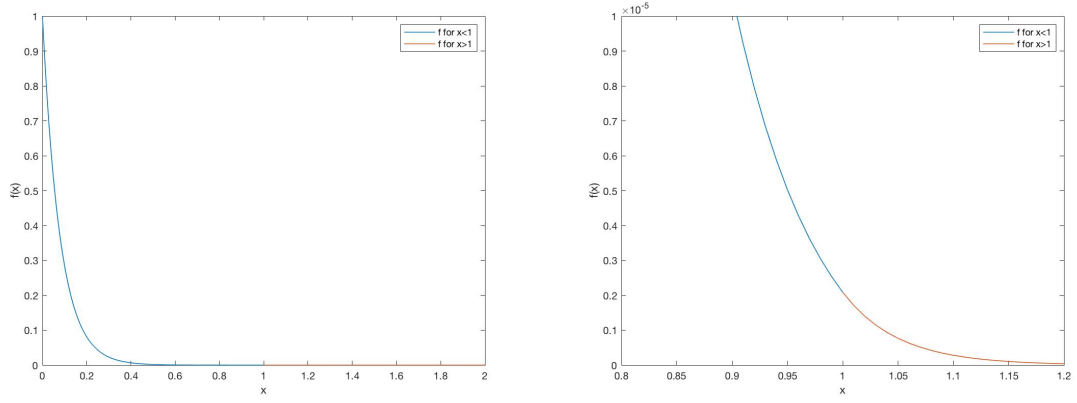


Figure 6: Graph of f as a function of x with $\hat{x} = 1$
(zoomed in to the right)

With K , μ and $\hat{x}$ as before, $f(x)$ is plotted for different values of K in Figure 7 below. It can be seen how the second moment gets smaller as K decreases, as expected.

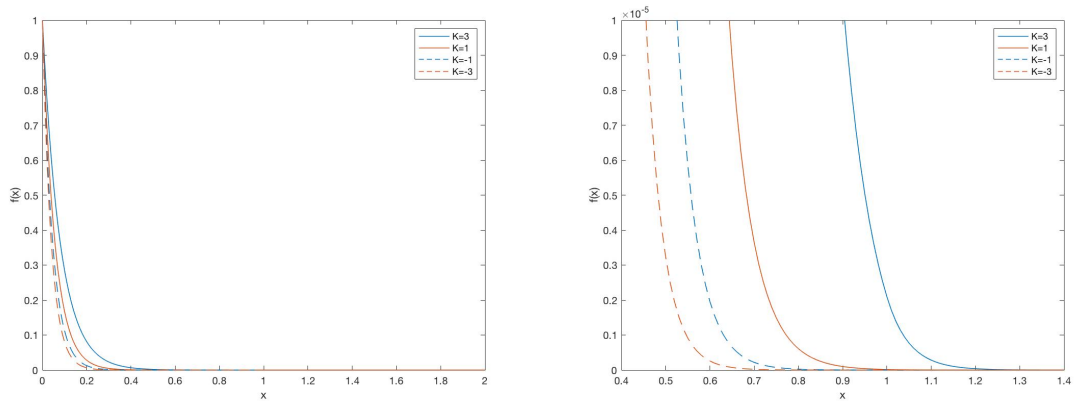


Figure 7: Graph of f as a function of x for different K
(zoomed in to the right)

6 Optimal Importance Sampling for Diffusions

Continuing on previous problems, when changing measure to minimize the variance of an estimation there might arise a computational cost from "extreme" drifts. Another approach than the importance sampling and Girsanov change of measure to change the drift of the diffusion process could be a stochastic control problem where we try to find the optimal new added drift $\hat{b}$. We want to formulate this problem as a Hamilton-Jacobi-Bellman equation with a penalty term that in some way reflects the computational issues that arise with this problem. The basic control problem of the second moment u with control constraint b denoting the new added drift will according to Theorem 4.1 be

$$\frac{1}{2}u_{xx} + \inf_b \{ (b + \mu)u_x + b^2u \} = 0.$$

In this set up, the solution will degenerate. By adding a penalty term, this issue is taken care of. There are several different options on penalty terms that may be considered, some examples could be

$$cb^2, \ c|b|, \ c|b|^3, \ c(b + \mu)^2 \text{ and } c(b')^2. \quad (21)$$

One option is to try to penalize large Girsanov changes and changes that will result in a diffusion process far from the original model. The first three options in equation (21) would all somehow reflect on that motive. The next item on the list penalizes the sum of the original and added drift being large and the last example would reflect a penalty for large derivatives of the change of the drift. This would be outside the framework of Theorem 4.1.

There are other penalties that possibly reflects the numerical problem better, but these might be hard to formulate. A continuation of this problem could be to investigate penalty terms reflecting the variance and fluctuation of the drift in the diffusion model under the new measure. An adapted method with smaller size of the time steps where the diffusion process fluctuates, and larger time steps where it behaves more flat and even, would take into account some important computational matters. Although, this would be too complicated to formulate in a stochastic control problem.

We choose to formulate this problem as the Hamilton-Jacobi-Bellman equation with penalty term b^2 that penalizes large changes of the drift, i.e., it penalizes large new drifts, and some constant $c \geq 0$. With the constant added drift b and with u being the second moment from equation (18), we have

$$\frac{1}{2}u_{xx} + \inf_b \{ (b + \mu)u_x + b^2u + cb^2 \} = 0 \quad (22)$$

and

$$u(x) = \inf_b \mathbb{E} \left[\mathbb{1}_{\tau_0 < \infty} \left(e^{\int_0^{\tau_0} b^2 dt} + c \int_0^{\tau_0} e^{\int_0^s b^2 dt} b^2 ds \right) \right].$$

Finding the infimum by setting the derivative to 0 results in the equation

$$u_x + 2bu + 2cb = 0$$

which gives us the optimal control law

$$\hat{b} = -\frac{u_x}{2(u+c)}.$$

Inserting the optimal control law $\hat{b}$ into the HJB equation (22), we get

$$\frac{1}{2}u_{xx} + \mu u_x - \frac{u_x^2}{2(u+c)} + \frac{uu_x^2}{2^2(u+c)^2} + \frac{cu_x^2}{2^2(u+c)^2} = 0$$

which can be simplified to

$$\frac{1}{2}u_{xx} + \mu u_x - \frac{u_x^2}{4(u+c)} = 0. \quad (23)$$

The next step is to solve the ordinary differential equation from equation (23) before inserting it into the optimal control law function. It can be linearized by the substitution $u+c = v^\gamma$. We get the following derivatives

$$u_x = \gamma v^{\gamma-1} v_x$$

and

$$u_{xx} = \gamma(\gamma-1)v^{\gamma-2}v_x^2 + \gamma v^{\gamma-1}v_{xx}.$$

Substituting accordingly gives us

$$\begin{aligned} \frac{1}{2}u_{xx} + \mu u_x - \frac{u_x^2}{4(u+c)} &= \frac{1}{2}(\gamma(\gamma-1)v^{\gamma-2}v_x^2 + \gamma v^{\gamma-1}v_{xx}) + \mu \gamma v^{\gamma-1}v_x - \frac{(\gamma v^{\gamma-1}v_x)^2}{4v^\gamma} \\ &= \gamma v^{\gamma-1} \left(\frac{\gamma-1}{2v} v_x^2 + \frac{1}{2}v_{xx} + \mu v_x - \frac{\gamma v_x^2}{4v} \right) \\ &= 0. \end{aligned}$$

We choose γ to make the v_x^2 -terms cancel out, hence we need

$$\frac{\gamma-1}{2v} = \frac{\gamma}{4v}$$

which gives us $\gamma = 2$. We are left with

$$\frac{1}{2}v_{xx} + \mu v_x = 0,$$

which has solution

$$v = C_0 + C_1 e^{-2\mu x}$$

for some $C_0, C_1 \in \mathbb{R}$. Thus

$$u = (C_0 + C_1 e^{-2\mu x})^2 - c.$$

We have the initial and boundary conditions $u(0) = 1$ and $u(\infty) = 0$ and we know that $0 \leq u \leq 1$ which implies $c \leq v^2 \leq c + 1$. This gives us that

$$u(0) = (C_0 + C_1)^2 - c = 1$$

and

$$u(\infty) = C_0^2 - c = 0.$$

Thus, $C_0 = \sqrt{c}$ and $C_1 = \sqrt{c+1} - \sqrt{c}$. It follows that

$$u = (\sqrt{c} + (\sqrt{c+1} - \sqrt{c}) e^{-2\mu x})^2 - c. \quad (24)$$

Inserting u into the optimal control law function, we get

$$\begin{aligned} \hat{b} &= -\frac{u_x}{2(u+c)} \\ &= -\frac{-2\mu(\sqrt{c+1} - \sqrt{c}) e^{-2\mu x} 2(\sqrt{c} + (\sqrt{c+1} - \sqrt{c}) e^{-2\mu x})}{2(\sqrt{c} + (\sqrt{c+1} - \sqrt{c}) e^{-2\mu x})^2} \\ &= \frac{2\mu(\sqrt{c+1} - \sqrt{c}) e^{-2\mu x}}{\sqrt{c} + (\sqrt{c+1} - \sqrt{c}) e^{-2\mu x}} \\ &= \frac{2\mu(\sqrt{c+1} - \sqrt{c})}{e^{2\mu x} \sqrt{c} + (\sqrt{c+1} - \sqrt{c})}. \end{aligned}$$

Remark. Revisiting Example 3.1, we know that the probability of reaching the level b in finite time is $e^{-2\mu b}$ for the process

$$X_t = -\mu t + W_t, \quad \mu > 0$$

with $X_0 = 0$, where W_t is a Brownian motion. Now we have a similar but reversed situation where we start at $X_0 = x \geq 0$ and wonder when the process hits 0. Probability wise, the process behaves the same. Thus, $p = e^{-2\mu x}$ and $p^2 = e^{-4\mu x}$.

When $c \rightarrow 0$, it appears $\hat{b} = 2\mu$ and $u = (e^{-2\mu x})^2$. With no penalty, i.e., when $c = 0$, we obtain that $u = p^2$, with p according to equation (16). This corresponds to zero variance which reflects the set up where importance sampling with the right new drift $\hat{b}$ generates a perfect estimate. However, such optimal importance sampling typically imposes exploding drifts, which results in complications when simulating trajectories. Hence, the penalty c would adjust the choice of optimal drift $\hat{b}$ to something that is good enough both estimation wise and implementation wise.

Remark. As previously discussed, looking at equation (24) we see that $u \rightarrow (e^{-2\mu x})^2$ as $c \rightarrow 0$. Note that since u represents the second moment, we need to have

$$u \geq p^2 = (e^{-2\mu x})^2 = e^{-4\mu x}$$

with p according to equation (16). If this would not be the case, it would imply that the variance is negative which cannot happen, and therefore the problem would be

badly formulated. To see that it actually holds for every $c \geq 0$, look at

$$\begin{aligned} u &= (\sqrt{c} + (\sqrt{c+1} - \sqrt{c}) e^{-2\mu x})^2 - c \\ &= (\sqrt{c+1} - \sqrt{c})^2 e^{-4\mu x} + 2\sqrt{c} (\sqrt{c+1} - \sqrt{c}) e^{-2\mu x} \\ &= (\sqrt{c+1} - \sqrt{c}) e^{-4\mu x} \left[(\sqrt{c+1} - \sqrt{c}) + 2\sqrt{c} e^{2\mu x} \right]. \end{aligned}$$

Thus, we need to show that the factor

$$u_{\text{factor}} = (\sqrt{c+1} - \sqrt{c}) \left[(\sqrt{c+1} - \sqrt{c}) + 2\sqrt{c} e^{2\mu x} \right] \geq 1,$$

for $u \geq e^{-4\mu x}$ to hold. We continue

$$\begin{aligned} u_{\text{factor}} &= (\sqrt{c+1} - \sqrt{c}) \left[(\sqrt{c+1} - \sqrt{c}) + 2\sqrt{c} e^{2\mu x} \right] \\ &= 1 + 2c - 2\sqrt{c}\sqrt{c+1} + 2\sqrt{c}(\sqrt{c+1} - \sqrt{c})e^{2\mu x} \\ &= 1 - 2\sqrt{c}(\sqrt{c+1} - \sqrt{c}) + 2\sqrt{c} e^{2\mu x}(\sqrt{c+1} - \sqrt{c}) \\ &= 1 + (\sqrt{c+1} - \sqrt{c})(2\sqrt{c} e^{2\mu x} - 2\sqrt{c}). \end{aligned}$$

For every $c \geq 0$, we have that $(\sqrt{c+1} - \sqrt{c}) > 0$ and further for every $\mu > 0$ and $x > 0$, we also have that $(2\sqrt{c} e^{2\mu x} - 2\sqrt{c}) > 0$. Hence,

$$u_{\text{factor}} \geq 1.$$

Note that this does not hold for all $x \in \mathbb{R}$. Though, in this set up we handle the problem of the first time a process X hits 0, starting at $X_0 = x > 0$. Thus, this only being valid for $x > 0$ covers the problem.

6.1 Other Penalty Terms

As explained in previous section, there are more than one penalty term that could describe our problem well. We have done a rigorous calculation of the second moment u and its optimal control law $\hat{b}$ for the penalty term cb^2 . The choice of cb^2 could have been replaced by for example any of the options in equation (21), although they might have resulted in more complicated solutions of the Hamilton-Jacobi-Bellman equation without actually being an improvement of the model as a whole.

Example 6.1. With the same set up as in Section 6, let instead the penalty term be $c|b|$. Then the HJB equation will become

$$\frac{1}{2}u_{xx} + \inf_b \{ (b + \mu)u_x + b^2u + c|b| \} = 0.$$

Similarly, after finding the infimum we would end up with

$$\begin{cases} u_x + 2bu + 2c = 0, & \text{for } b > 0, \\ u_x + 2bu - 2c = 0, & \text{for } b < 0, \end{cases}$$

and the optimal control law

$$\hat{b} = -\frac{u_x \pm 2c}{2u}$$

with $+2c$ for $b > 0$ and $-2c$ for $b < 0$. After replacing b with $\hat{b}$ in the original HJB equation, we get

$$\frac{1}{2}u_{xx} + \mu u_x + \left(\frac{u_x}{2u}\right)^2 (1 - 2u) + \frac{u_x}{2u^2}(2c - u(2 + c)) + \frac{c^2}{2u^2}(2 - u) = 0.$$

Thus, we end up with an ODE to solve for u . We leave the details of the calculations.

Example 6.2. Continuing in the same manner but with penalty term $c(b + \mu)^2$, the HJB equation will become

$$\frac{1}{2}u_{xx} + \inf_b \{(b + \mu)u_x + b^2u + c(b + \mu)^2\} = 0.$$

The optimal control law would become

$$\hat{b} = -\frac{u_x + 2c\mu}{2(u + c)}.$$

Again, we will end up with an ODE to solve for u , just as in Example 6.1.

Remark. If we would instead approach the same calculations with the penalty term $c(b')^2$, given that b is not a constant, we arrive at a more difficult problem depending on the attributes of the new added drift function b . A general example of this sort is not possible for us to calculate, but with constant drifts b as in Example 6.1 and 6.2, it is possible to calculate a general solution for the second moment u in the HJB equation. As pointed out before, this would be outside the framework of Theorem 4.1. Again, there are numerous ways to account for numerical issues in this problem and hence several possible penalty terms. The example that we chose to look further at in Section 6, we chose to prosecute with since it results in a relatively easy version of the HJB equation while it still takes into account the desired penalization of large changes in the drift.

6.2 The Finite Horizon Version

These kinds of problems that have been treated so far in this thesis can be formulated in many different ways depending on what question you would like to answer. In the previous sections about stochastic optimal control problems we have translated the question of when the first time a stochastic process hits zero. Before that, in Section 5.1, we were interested in the probability that the process X is larger than a certain non-negative barrier B at time T . Problems of this sort would result in a finite version of the Hamilton-Jacobi-Bellman equation with an extra term u_t according to

$$u_t + \frac{1}{2}u_{xx} + \inf_b \{(b + \mu)u_x + b^2u + cb^2\} = 0.$$

Again, we would get the optimal control law

$$\hat{b} = -\frac{u_x}{2(u+c)},$$

but when inserting $\hat{b}$ into the HJB equation we would instead get a partial differential equation

$$u_t + \frac{1}{2}u_{xx} + \mu u_x - \frac{u_x^2}{4(u+c)} = 0.$$

We have not been able to obtain a closed-form solution of this non-linear equation.

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